

29 Matrix Kth smallest element in a row-column wise sorted matrix

```
#include <bits/stdc++.h>
using namespace std;
class Solution {
public:
    int kthSmallest(vector<vector<int>> &mat, int k) {
        int n = mat.size();
        if (n == 0) return -1;
        int m = mat[0].size();
        if (m == 0) return -1;
        // Min-heap of tuples: (value, row, col)
        using T = tuple<int,int,int>;
        auto cmp = [](const T &a, const T &b){
            return get<0>(a) > get<0>(b);
        };
        priority_queue<T, vector<T>, decltype(cmp)> pq(cmp);
        // Push first element of each row (if exists)
        for (int r = 0; r < n; ++r) {
            if (!mat[r].empty())
                pq.emplace(mat[r][0], r, 0);
        }
        int val = -1;
        for (int i = 0; i < k; ++i) {
            if (pq.empty()) break; // defensive
            auto [v, r, c] = pq.top();
            pq.pop();
            val = v;
            // push next element in the same row
            if (c + 1 < (int)mat[r].size()) {
                pq.emplace(mat[r][c+1], r, c+1);
            }
        }
        return val;
    }
};
```

The screenshot displays a coding platform interface with two main panels. The left panel, titled 'Output Window', shows the 'Compilation Results' for a submission by 'Y.O.G.J. (AI Bot)'. It indicates that the 'Problem Solved Successfully' with 1120 test cases passed out of 1120, an accuracy of 100%, 4 out of 4 points scored, and a time taken of 0.45 seconds. Below this, there are buttons for 'Solve Next' and 'Stay Ahead With:' followed by links to other problems like 'Find kth element of spiral matrix'. The right panel shows the source code of the solution, which is a C++ implementation using a min-heap (priority_queue) to find the kth smallest element in a row-column wise sorted matrix. The code is identical to the one provided in the text above.